Network Security Essentials – SOC Investigation Report
(TryHackMe Lab)

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Lab : TryHackMe —Network Security Essentials
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Executive Summary

This report presents a Security Operations Center (SOC) investigation conducted using Splunk SIEM within the *Network Security Essentials* lab. The objective was to analyze network logs, identify suspicious behavior, and demonstrate detection techniques using real-world SOC workflows.

The investigation progressed from general log visibility to targeted threat detection by filtering specific log sources and identifying blocked network activity. Findings indicate the presence of suspicious traffic patterns consistent with reconnaissance or unauthorized access attempts.

1. Introduction

Security Operations Centers rely on SIEM platforms to aggregate and analyze logs from multiple sources. This lab simulates a real SOC workflow where analysts move from broad visibility to focused threat detection.

The investigation demonstrates how analysts:

- Explore raw network data
- Filter logs by source type
- Detect blocked or malicious activity

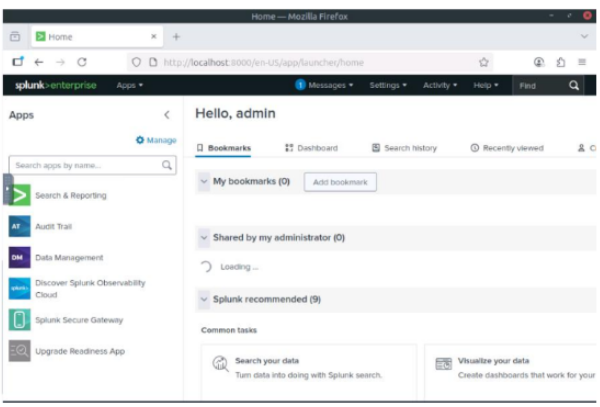
2. Technical Environment

Component	Description
SIEM Platform	Splunk Enterprise
Data Source	Network logs
Log Types	Firewall logs
Analysis Type	Log-based threat detection

3. Initial SIEM Access & Visibility

SOC Analysis:
The Splunk dashboard provides centralized access to all ingested logs. This confirms that log collection is active and that the SOC has visibility into network activity.

From a SOC perspective, this is the **visibility layer**, which is critical before any investigation begins.



4. Initial Log Retrieval

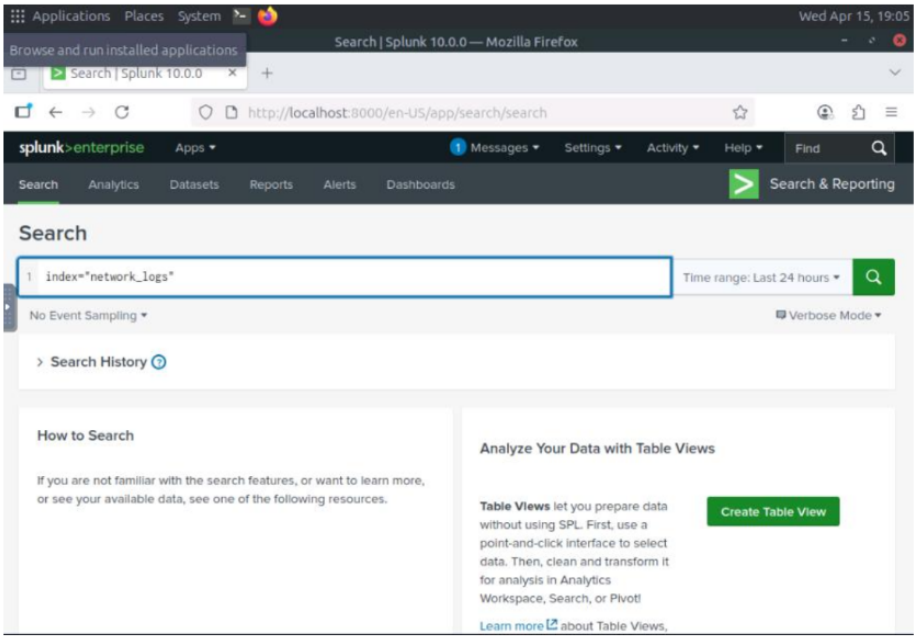
Query used:

```
index="network_logs"
```

SOC Analysis:
This query retrieves all network-related logs. At this stage, the analyst is performing **broad data exploration** to understand the dataset.

SOC Insight:

> ***Always start wide, then narrow down.***



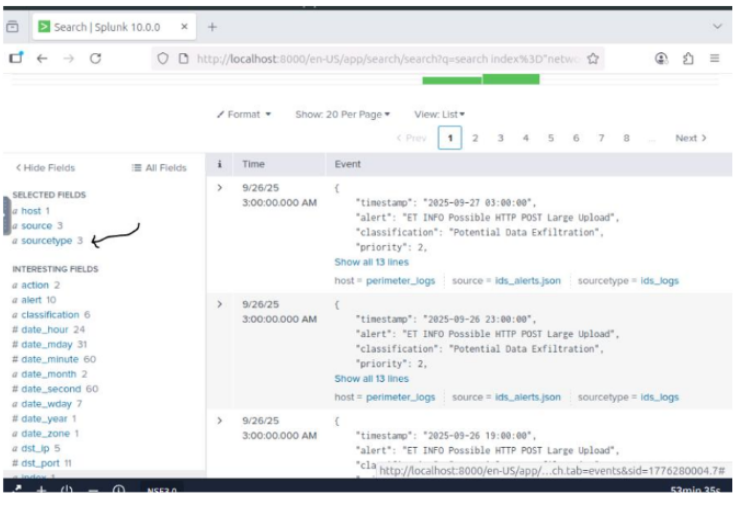
5. Log Source Identification (Sourcetype Analysis)

The next step involved identifying specific log types using the **sourcetype field**.

SOC Analysis:
Filtering by *sourcetype* allows the analyst to focus on specific data sources. In this case, the analyst identified **firewall logs** as a high-value source for detecting malicious activity.

SOC Insight:
Firewall logs are one of the most critical data sources for detecting:

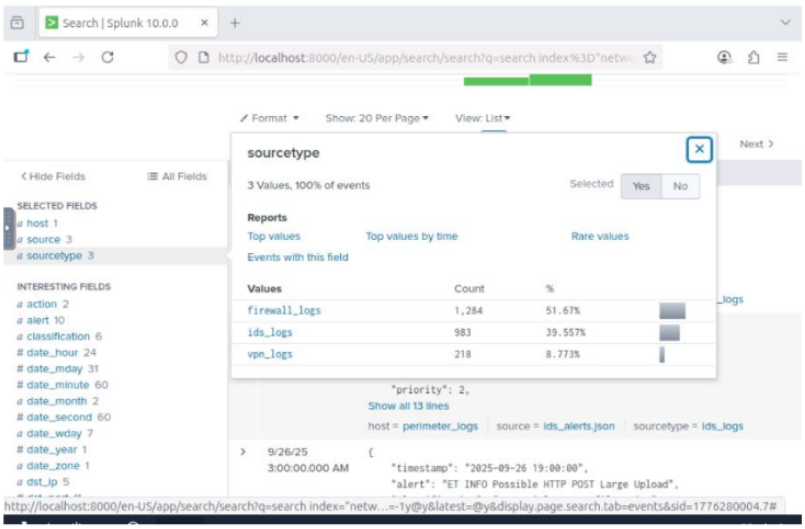
- Unauthorized access attempts
- Blocked connections
- External threats



6. Focused Analysis – Firewall Logs

The investigation was narrowed to:

```
sourcetype="firewall_logs"
```



SOC Analysis:
By isolating firewall logs, the analyst reduces noise and focuses on **network perimeter activity**, where most attacks originate.

SOC Interpretation:
This step transitions the investigation from **general monitoring → targeted threat analysis**.

7. Detection of Blocked Activity

Final query used:

```
sourcetype="firewall_logs" action=block
```

SOC Analysis:
This query identifies all network events where the firewall actively **blocked traffic**.

Key Findings:

- Presence of blocked connection attempts
- Evidence of denied access from external sources
- Indicators of potential malicious activity

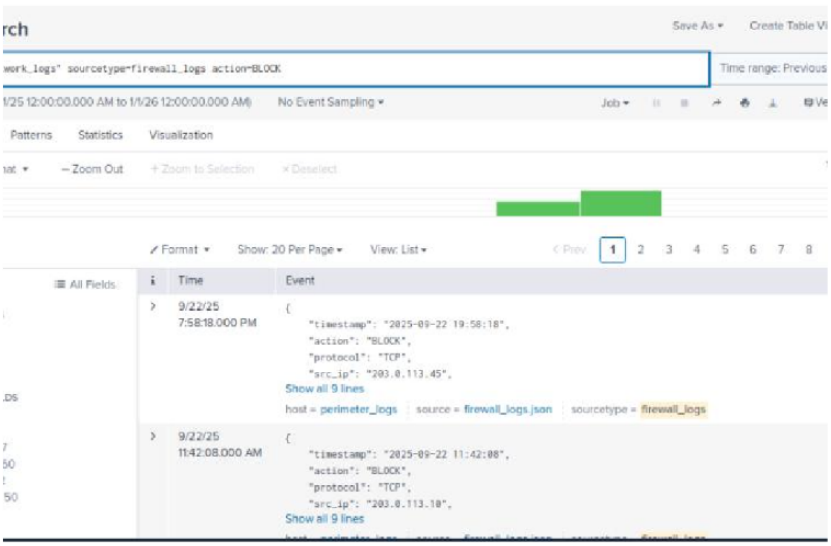
SOC Interpretation:

Blocked traffic is a **strong indicator of suspicious behavior**, often associated with:

- Port scanning attempts
- Unauthorized access attempts
- Malicious external probes

SOC Insight:

Firewalls don't block normal behavior — they block suspicious or policy-violating traffic.



8. Advanced Threat Hunting (SIEM Analysis)

To extend the investigation beyond firewall analysis, additional threat-hunting queries were performed on VPN and IDS logs using Splunk SPL.

8.1 VPN Log Analysis

Queries used:

```
index="network_logs" sourcetype="vpn_logs"
index="network_logs" sourcetype="vpn_logs" result="SUCCESS"
index="network_logs" sourcetype="vpn_logs" result="SUCCESS" username="svc_backup"
```

SOC Analysis:

The investigation focused on successful VPN authentication events, which are often overlooked compared to failed login attempts. The identification of successful logins for the account **"svc_backup"** is particularly significant.

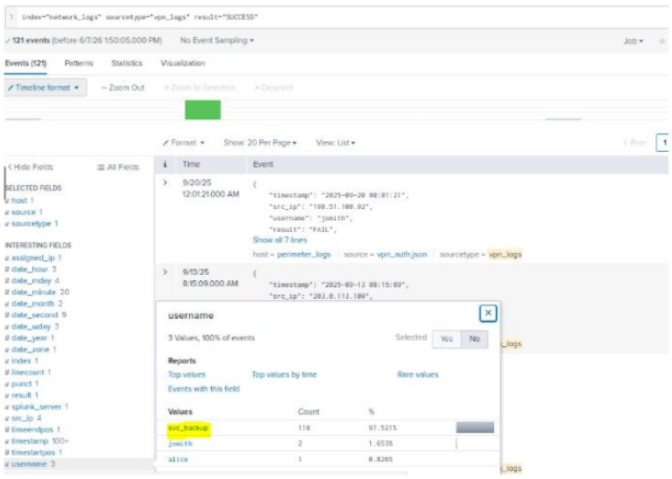
SOC Interpretation:

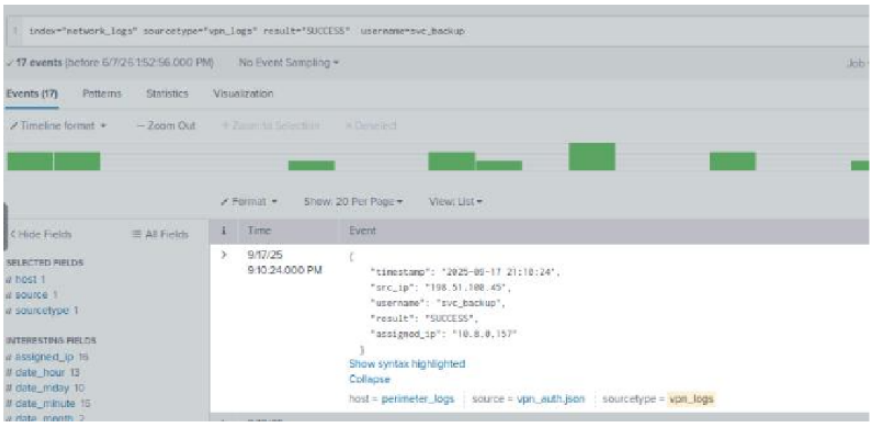
- Service accounts are typically used for automated processes and should have restricted access patterns
- Unexpected or unusual login activity involving service accounts may indicate compromise
- Successful authentication does not imply legitimacy

Potential Risk:

This activity may represent:

- Unauthorized access using valid credentials
- Credential compromise
- Abuse of service accounts for persistence





8.2 IDS Alert Analysis

Queries used:

```
index="network_logs" sourcetype=ids_logs
index="network_logs" sourcetype=ids_logs classification="Potential Data Exfiltration"
index="network_logs" sourcetype=ids_logs alert="ET TROJAN Possible C2 Beaconing"
index="network_logs" sourcetype=ids_logs classification="Attempted Unauthorized Access"
```

SOC Analysis:

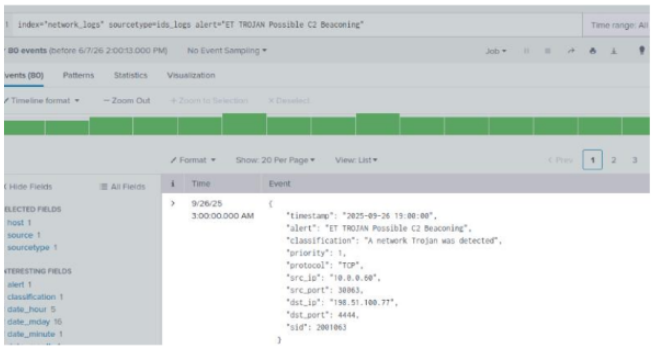
IDS logs revealed multiple high-risk alerts, including:

- Potential Data Exfiltration
- Command-and-Control (C2) Beaconing
- Unauthorized Access Attempts

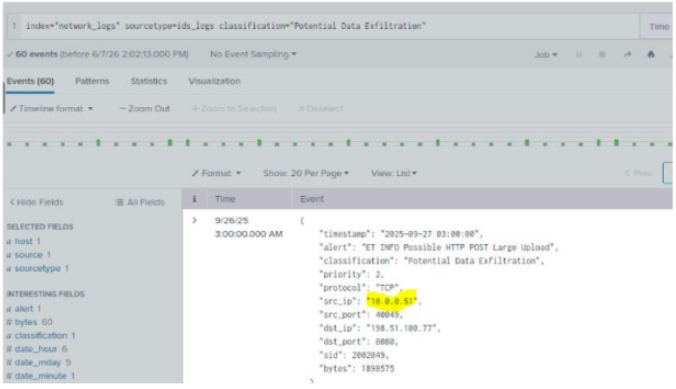
SOC Interpretation:

These alerts indicate strong signs of malicious activity:

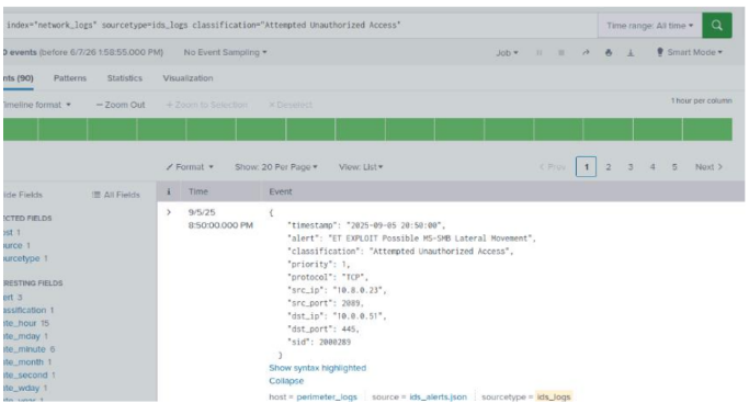
- **C2 Beaconing** suggests communication with an attacker-controlled server



- **Data Exfiltration** alerts indicate possible sensitive data transfer



- **Unauthorized access attempts** reinforce intrusion activity



SOC Insight:

> IDS alerts combined with SIEM correlation significantly increase confidence in detecting real threats.

8.3 Threat Correlation

When correlating VPN, firewall, and IDS data:

- Firewall logs show blocked suspicious traffic
- VPN logs show successful access events
- IDS logs indicate malicious behavior patterns

SOC Conclusion:

This combination strongly suggests a **potential attack chain**, where:

- Initial access may have been achieved via VPN
- Internal activity triggered IDS alerts
- Firewall enforced partial protection

Recommended Immediate Actions

- Investigate the "svc_backup" account activity
- Reset credentials and enforce MFA
- Identify source IP of VPN connections
- Block any C2-related IPs/domains
- Perform endpoint investigation

Final SOC Insight

This phase of the investigation demonstrates the transition from basic monitoring to **true threat hunting**, where multiple data sources are correlated to uncover deeper indicators of compromise.

9. Threat Analysis & Behavioral Assessment

The investigation reveals:

- Repeated blocked connections
- External interaction attempts
- Network-level enforcement by firewall

SOC Interpretation:

These indicators strongly suggest:

- Reconnaissance activity (network scanning)
- Unauthorized access attempts
- Automated probing behavior

The structured nature of these events indicates potential scripted or automated activity rather than legitimate user behavior.

10. MITRE ATT&CK Mapping

Behavior	Technique	ID
Blocked connection attempts	Network Service Discovery	T1046
External probing	Reconnaissance	TA0043
Suspicious traffic	Initial Access	T1190

11. Detection Methodology

This investigation followed a real SOC workflow:

- Broad log search
- Identify relevant data source (sourcetype)
- Focus on firewall logs
- Filter by security-relevant actions (block)
- Analyze patterns and behavior

SOC Insight:
This layered approach is fundamental in real-world threat hunting.

12. Response Recommendations

Immediate Actions

Investigate source IP addresses generating blocked traffic
Apply rate-limiting or geo-blocking if necessary

Short-Term Improvements

Create Splunk alerts for repeated block events
Monitor high-frequency blocked attempts

Long-Term Strategy

Correlate firewall logs with IDS/IPS alerts
Integrate SIEM with threat intelligence feeds
Build automated detection rules

13. Final SOC Assessment

The investigation demonstrates a clear progression from visibility to detection using SIEM tools.

The identification of blocked firewall activity provides strong evidence of:

- Suspicious external interaction
- Potential reconnaissance or attack attempts

Final Verdict:

The activity is classified as **suspicious**, with indicators consistent with early-stage network attacks. While the firewall successfully mitigated the threat by blocking traffic, continued monitoring and correlation are required to fully assess the risk.

14. Conclusion

This lab highlights the importance of structured log analysis in SOC operations. By progressively refining queries and focusing on high-value data sources, analysts can efficiently detect potential threats.

The ability to:

- Filter logs effectively
- Identify meaningful patterns
- Interpret blocked activity is essential for any SOC analyst.

This investigation reflects real-world SOC practices and demonstrates how even basic SIEM queries can uncover critical security insights.